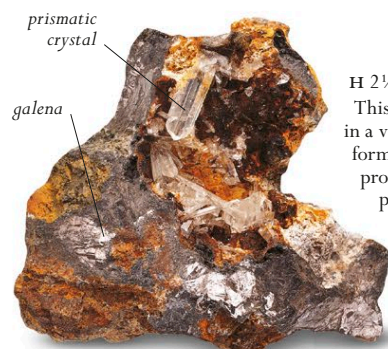
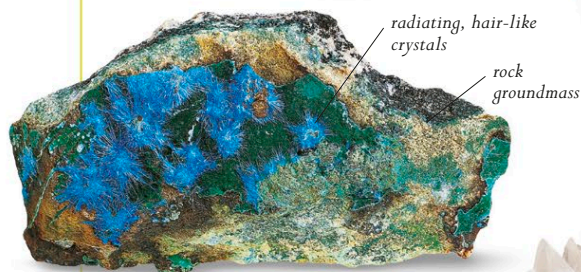


SULFATES

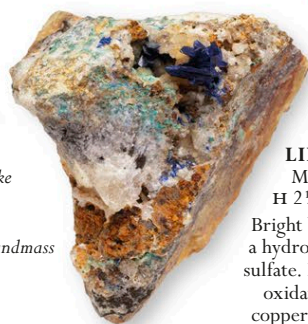
Sulfates are composed of metals combined with the sulfate radical (SO_4). There are about 200 sulfates and many are rare. Some, such as gypsum, form in evaporite deposits where minerals are precipitated from drying saline solutions. Others occur as weathering products, or as primary minerals in hydrothermal veins. Many are economically important – barite is used to lubricate drills on oil rigs.



ANGLESITE
Orthorhombic
H $2\frac{1}{2}$ –3 • SG 6.32–6.39
This lead sulfate occurs in a variety of colours and forms. It is an alteration product of galena, the primary lead ore.



CYANOTRICHITE
Monoclinic • H 1–3 • SG 2.76
This hydrated copper aluminium sulfate is named after the Greek for “blue” and “hair”, referring to its clusters of fine, blue crystals.



LINARITE
Monoclinic
H $2\frac{1}{2}$ • SG 5.35
Bright blue linarite is a hydrous copper lead sulfate. It occurs in the oxidation zones of copper and lead ores.



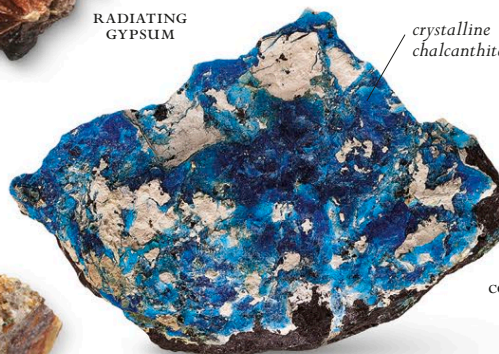
GLAUBERITE
Monoclinic
H $2\frac{1}{2}$ –3 • SG 2.75–2.85
Glauberite is a sodium and calcium sulfate. Colourless, grey, or yellowish, it forms where saline water evaporates.



RADIATING GYPSUM



GYPSUM
Monoclinic • H 2 • SG 2.31–2.32
A widespread mineral, gypsum, a hydrated calcium sulfate, produces plaster of Paris when heated and mixed with water.



crystalline chalcantithite

SATIN SPAR



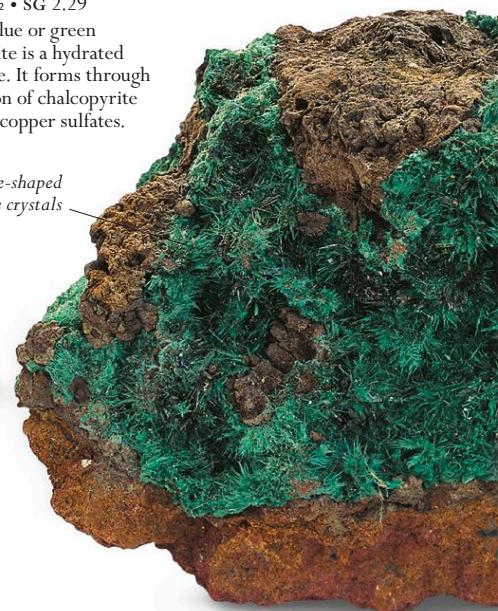
THENARDITE
Orthorhombic
H $2\frac{1}{2}$ –3 • SG 2.66
A pale grey or brownish mineral, thenardite is a sodium sulfate. It is found on lava flows and around salt lakes.

CHALCANTHITE
Triclinic
H $2\frac{1}{2}$ • SG 2.29
Rich blue or green chalcantithite is a hydrated copper sulfate. It forms through the oxidation of chalcocopyrite and other copper sulfates.



mass of needle-shaped brochantite crystals

ALUNITE
Trigonal
H $3\frac{1}{2}$ –4 • SG 2.6–2.9
A hydrous sulfate of potassium and aluminium, alunite may be found at volcanic vents where rocks are altered by sulfur vapours.

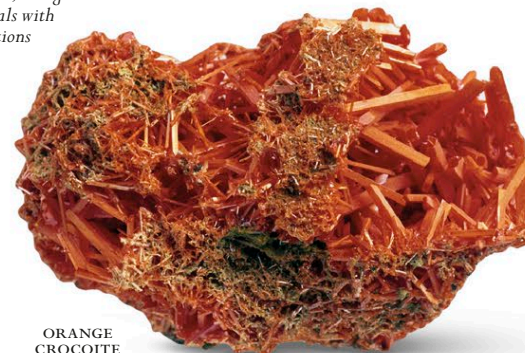


RED CROCOITE

CHROMATES

Chromate minerals form when metallic elements combine with the chromate radical (CrO_4). They are rare minerals – crocoite is the only reasonably well-known chromate. Generally brightly coloured, they are sought after by mineral collectors. Chromates often form when hydrothermal veins are altered by fluids.

slender, elongate crystals with striations



ORANGE CROCOITE

CROCOITE
Monoclinic • H $2\frac{1}{2}$ –3 • SG 5.97–6.02
Orange or red lead chromate forms in the oxidized zones of lead deposits. Good specimens come from Australia.